Software Requirements Specification

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Team name	Apollo
Project title	TravelM — Pal for Trip & Travel assistant
SDLC chosen in M1 + reason	Agile/Scrum — allows the team to develop the system iteratively, gather feedback, and continuously improve features throughout the project.
Date / version	v2.0 · 24 Sep 2026
GitHub repo	https://github.com/Albert69-se/Travel_M

1. Introduction

1.1 Purpose and scope

Travelers often manage destinations, attractions, weather information, packing lists, and trip expenses across different apps, websites, notes, and messages, making trip planning difficult and increasing the risk of relying on incomplete or outdated information. TravelM is a small travel assistant that brings these essential travel needs into one place, helping users discover and search for places, view accurate and sourced destination information, check weather conditions, organize itineraries, prepare packing lists, and track their travel budget and expenses. The MVP focuses on the core travel planning process, including discovering and filtering places, viewing detailed and verified place information, checking weather, creating and organizing trips, managing packing lists, and tracking expenses against a trip budget.

Success looks like this: a traveler can plan and manage the important parts of a trip in one place while accessing reliable, clearly sourced destination information without having to switch between multiple apps.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Traveler/ Tourist	primary user	Explore destinations and attractions, plan trips, check weather, manage bookings, packing lists, budgets, and expenses.	Inaccurate information, losing trip data, unexpected expenses, and poor usability.
Project Owner	Business stakeholder	A useful, reliable, secure, and maintainable Travel M application that meets the project's objectives and user needs.	Project failure, poor user adoption, security problems, excessive complexity, and failure to meet requirements.
System Administrator	Operational stakeholder	Manage users, system content, attraction information, categories, and system settings.	Unauthorized access, data corruption, system misuse, and security breaches.
Guest visitor	Optional secondary	Browse public attractions and weather information	Giving out personal information/private data

2. Overall description

2.1 Product context — what it does / does not

TravelM helps users discover places, plan trips, check destination weather, prepare packing lists, and track travel budgets and expenses. Everything described in the Functional Requirements is the whole MVP for this term — nothing else is implied.

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• FR-1 Discover places • FR-2 Search and filter places • FR-3 View accurate and sourced place information • FR-4 View weather for destination and trip dates • FR-5 Create and maintain a trip • FR-6 Set a budget for each trip • FR-7 Organize an itinerary by trip day • FR-8 Prepare and maintain a packing checklist • FR-9 Record and manage expenses • FR-10 View expense summary, status, reminder	• Hotel, flight, or restaurant booking • Payments or subscriptions • Public reviews, comments, or social networking • Complex AI recommendations • Real-time transport optimization

2.2 Assumptions and constraints

- Assumption: Every traveler has a mobile phone and access to the internet.
- Constraint: Place information must be sourced, accurate, and kept up to date where applicable.
- Constraint: Efficient and accurate calculations; calculations must be efficient and correct without restraining the system.
- Constraint: Personal data — we store only planned/saved trips, marked locations, username, and email address.

3. Functional requirements

FR-1 — Discover places

User story: As a tourist, I want to explore and view a list of popular tourist attractions..

Pain point: Popular places are often scattered on social media and many other platforms; it is hard to find a single list of popular places.

1. Once the tourist enters the website, they may view popular places on interactive maps.
2. Given the ability to mark the map, the tourists may mark and view the information about the place.

FR-2 — Search & Filter places

User story: As a tourist, I want to shrink the list by searching for keywords and filtering out only the ones I want.want.

Pain point: tourists see the whole map and are overwhelmed by the large number of places.

1. Given the search bar, tourists can search for attractions using keywords.
2. Tourists may also filter attractions by location or destination.
3. The system shall allow tourists to filter attraction by category, such as temples, nature, museums, markets, and viewpoints.

FR-3 — View accurate and sourced place information

User story: As a person who is still in the middle of trip planning, I want detailed and reliable information about a place beforehand so that I can make informed travel plans.

Pain point: tourists may encounter incomplete, outdated, or inaccurate information about places.

1. The system shall allow tourists to view detailed information about an attraction.
2. The system shall display the attraction location using Google Maps data.
3. The system shall display historical or cultural information about an attraction when available.
4. The system shall display useful information such as opening hours, entrance fees, and travel tips when available.
5. The system shall identify the source of curated place information and record the last-checked date where applicable.
6. Information that cannot be verified or may be outdated shall not be presented as verified/current information.

FR-4 — View weather for destination and trip dates

User story: As a tourist, I want to know the normal temperature and rainfall of the selected attraction, and the weather forecast. of the selected attraction, and the weather forecast.

Pain point: tourists often have trouble packing clothes and useful items with very little information about the weather.

1. The system shall allow tourists to view the current weather and temperature of selected destinations.
2. The system shall provide weather forecasts for tourist's planned travel dates.
3. The system shall display relevant weather information such as temperature, rainfall, and other weather conditions including storms and other incidents.
4. The system shall associate weather information with planned destinations and activities when applicable.

FR-5 — Create and maintain a trip

User story: As a traveler, I want to create, edit, or delete a trip with a name, date range, destinations, and financial record, so that each journey has a clear planning container. so that each journey has a clear planning container.

Pain point: Trip details become mixed when they are stored in general notes.

1. Given a name, valid date range, at least one destination, and activities when the traveler saves, then one trip is created with those values.
2. Given an existing trip, when the traveler edits it, then the same trip is updated; when deletion is confirmed, dependent records follow the approved deletion rule.

FR-6 — Set a budget for each trip

User story: As a traveler, I want to set or record an estimate budget for the trip, so that spending is evaluated against the correct amount.

Pain point: A general calculator cannot reliably connect totals, record the expense, and remind the spent on a specific trip.

1. Given a non-negative amount, when the traveler saves the budget, then that amount becomes the trip's expected budget.
2. When the traveler adds a new expense, then the total expected budget is unchanged and the remaining budget is recalculated accurately at once.

FR-7 — Organize and itinerary by trip day

User story: As a traveler, I want to add, remove, and order places or activities by trip day, so that the itinerary is understandable before departure.

Pain point: A saved-place list does not show what should happen each day.

1. Given a trip item and a valid trip day, when the traveler adds it, then it appears under that day with a stable order.
2. Given an itinerary item, when the traveler moves or removes it, then the affected day lists update without deleting the underlying place record.

FR-8 — Prepare and maintain a packing checklist

User story: As a traveler, I want an editable packing checklist informed by trip context, so that I can prepare without relying on a separate note.

Pain point: Packing lists are frequently generic and disconnected from destination, weather, and planned activities.

1. Given a trip, when the traveler creates a packing list, suggested items may be drafted for trip dates, destinations, activities, and possible weather, and every suggestion remains editable or removable.
2. Given the traveler's checks, adds, renames, reorders, or removes an item, when the change is saved, then the packing list preserves the traveler-confirmed state rather than regenerating it.

FR-9 — Record and manage expenses

User story: As a traveler, I want to add, edit, delete, and categorize expenses, so that trip spending remains complete and correct.

Pain point: Expenses in notes require repeated manual addition and are easy to omit or mistype.

1. Given a positive amount in the trip budget, allowed category, date, and optional description, when the traveler saves, then that expense is only linked to the affected trip.
2. Given an existing expense, when it is edited or deletion is confirmed, then the same record is updated or removed, and no duplicate stays.

FR-10 — View expense summary, status and reminder

User story: As a traveler, I want to see totals, category groups, remaining budget, and clear warnings, so that I can understand and control trip spending.

Pain point: Separate notes and calculators make it easy to rely on an old total or notice overspending too late.

1. Given a trip's expected budget of 1,000 and expenses totaling 235, when the summary opens, then it shall show an expected budget of 1,000, total expenses of 235, and remaining budget of 765.
2. Given expenses exist, when the summary opens, then total spending shall also be grouped by the approved expense categories without changing the underlying expense records.
3. Given total spending reaches 80% or more of the expected budget, when the summary is displayed, then a ‘nearing budget warning’ shall be shown.
4. Given total spending exceeds the expected budget, and the remaining budget is less than 0, when the summary is displayed, then an ‘over-budget warning’ shall be shown.

4. Non-functional requirements

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	Place's detail view, keyword search with location and category filters, weather information and total budget and warnings shall finish loading/updating within ~3-5 seconds	Run and record the slowest results from - three keyword searches and three combined filter checks.filter checks. - three requests for three destinations and record the slowest successful result. requests for three destinations and record the slowest successful result. - visible update time in three trials for each operation.
NFR-2	Usability	Explore places, create a trip, add one place, add an itinerary item/place to a trip, add one packing item, set a budget, and record one expense within 5 minutes without help and within no more than three navigation actions from the home or trip overview.	Timed task with at least three representative travelers who have not seen the design.
NFR-3	Compatibility	Required content shall remain readable for both mobile, tablet and laptop views in current supported Chrome, Safari, Firefox, and Edge releases	Run the agreed responsive/browser matrix and record failures.
NFR-4	Security and privacy	An authenticated user shall be unable to read or modify another user's trips, packing items, expenses, or saved places by changing an identifier. Guest users are limited to only browsing/viewing places and weather information.	Attempt cross-user reads and writes for every owned entity and confirms rejection.
NFR-5	Input integrity	The system shall reject invalid date ranges, unsupported currency codes, non-positive expenses, out-of-range coordinates, invalid itinerary days, and incomplete required booking fields.	Submit one invalid case for each rule and verify that no invalid record is stored.
NFR-6	Reliability	Successfully saved trips, itinerary items, booking references, packing items, budgets, expenses, and saved-place links shall remain after application restart and a new authenticated session.	Save one of each record type, restart, sign in again, and compare the records.
NFR-7	External-service resilience	If a map/place, weather is unavailable, the system shall show a source-specific message and shall not corrupt or delete Travel_M-owned or user-owned records.	Simulate each provider's failure and verify the message and local records.
NFR-8	Calculation accuracy	Remaining budget and expenses are to be calculated accurately using 'Remaining Balance = Total Budget – Total expenses'	Test fixed decimal values, an overspent case, and add/edit/delete operations against an independent calculation.
NFR-9	Information accuracy	TravelM-curated place information shall have an identifiable and appropriate source and a last-checked date where applicable. Information that is outdated or cannot be verified shall not be presented as verified/current.	Review a sample of place records and verify that each has a source and last-checked date where applicable. Compare the displayed information with its referenced source and check that outdated/unverifiable information is appropriately marked or excluded.
NFR-10	Maintainability and provenance	Place, trip, weather, booking, packing, and budget concerns shall be separated into maintainable modules; map/weather	Review module boundaries, replace one provider with a test adapter, run contract tests, and independently update one curated record.

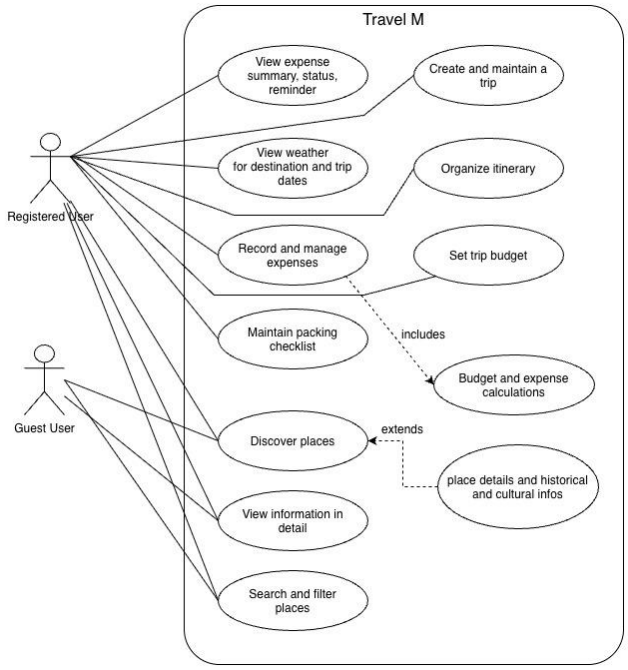
ID	Kind	Statement (testable)	How we will check
		providers shall be replaceable through adapters; curated records shall retain source, last-checked date, and status.	
NFR-11	Scalability	In a representative test deployment, 50 concurrent users shall complete the core read/write scenario with less than 1% request errors and a 95th-percentile owned-data response time of ~3-5 seconds or less, excluding external-provider latency.	Run a scripted 50-user load test and retain the report.
NFR-12	Availability	After a production environment and monitoring method exists, TravelM's owned functions shall achieve at least 99% monthly availability, exclude announced maintenance and separately report external-provider outages.	Calculate monthly availability from independent uptime checks and incident records.

5. Use cases

5.1 Use-case list

	Use case	Actor	Related FR	Brief success story
UC-01	Discover places	Registered user, Guest user	FR-1	Views relevant places on a list and map
UC-02	Search and filter places	Registered user, Guest user	FR-2	See available providers and curated information with provenance/staleness behavior.
UC-03	View information in detail	Registered user, Guest user	FR-3	See a place's details, historical, cultural and background stories if there's any available
UC-04	View weather for destination and trip dates	Registered user	FR-4	Available time- and source-qualified weather is shown for trip context.
UC-05	Create and maintain a trip	Registered user	FR-5	A dated, destination-aware, single-currency trip is created, edited, or deleted.
UC-06	Set trip budget	Registered user	FR-6	One expected budget in the trip currency is retained.
UC-07	Organize itinerary	Registered user	FR-7	Places or activities are ordered by valid trip day.
UC-08	Maintain packing checklist	Registered user	FR-8	Editable packing items are retained, optionally informed by available weather and activities.
UC-09	Record and manage expenses	Registered user	FR-9	Categorized expense records are added, edited, or deleted.
UC-10	View expense summary, status, reminder	Registered user	FR-10	Totals, category groups, remaining budget, and deterministic warnings are shown.

5.2 Use-case diagram



6. Out of scope

1. Hotel, flight, or restaurant booking
2. Payments or subscriptions
3. Public reviews, comments, or social networking
4. Complex AI recommendations
5. Real-time transport optimization

7. Traceability matrix (Golden Thread)

Only FR-3 is filled all the way across — that is the single thread this course follows week by week (Ch3 → Ch4/5 → Ch6 → Ch7). The other rows stay open until their week arrives, exactly like your own M2 should.

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Popular places are scattered across social media and many other platforms, making them difficult to find in one place	FR-1 Discover places	(Week 5)	(Week 7)	(Week 8)
Tourists can become overwhelmed by seeing many places and need a way to narrow their choices.	FR-2 Search & Filter places	(Week 5)	(Week 7)	(Week 8)
Tourists may receive incomplete or inaccurate information about places they plan to visit.	FR-3 View accurate and sourced place information	Provide detailed, sourced, and checked place information, with outdated or unverifiable information clearly excluded or marked	(Week 7)	(Week 8)

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Tourists may have difficulty preparing suitable clothing and useful items because they lack weather information for their destination and travel dates.	FR-4 View weather for destination and trip dates	(Week 5)	(Week 7)	(Week 8)
Trip details become mixed and difficult to manage when stored in general notes.	FR-5 Create and maintain a trip	(Week 5)	(Week 7)	(Week 8)
A general calculator cannot reliably connect a budget and expenses to a specific trip.	FR-6 Set a budget for each trip	(Week 5)	(Week 7)	(Week 8)
A saved place list does not show what should happen on each day of the trip.	FR-7 Organize and itinerary by trip day	(Week 5)	(Week 7)	(Week 8)
Packing lists are often generic and disconnected from destination, weather, and planned activities.	FR-8 Prepare and maintain a packing checklist	(Week 5)	(Week 7)	(Week 8)
Expenses recorded in notes require repeated manual addition and are easy to omit or mistype.	FR-9 Record and manage expenses	(Week 5)	(Week 7)	(Week 8)
Separate notes and calculators make it easy to rely on an old total or notice overspending too late.	FR-10 View expense summary, status and reminder	(Week 5)	(Week 7)	(Week 8)

8. AI usage log

If this table is empty but the prose looks generated, the milestone is capped at 50% — log every prompt that drafted a story or paragraph, honestly.

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
25 Aug 2026	Claude	Maintenance of the app	Got a suggestion that we should build the apps with different modules that are easy to maintain and scale as the app grows in the future	We changed the idea from building a whole thing together as one to using modules (locational and weather data, budget, user accounts)	Modularity and scalability are important factors in building and maintaining a good app
27 Aug 2026	Gemini	Incorporating advanced features (including connection with TripAdvisor, Booking.com and other ticketing/booking platforms)	Proposed a full architecture for connecting those platforms with various adapters	We dismissed the idea of it being too complicated and might possibly affect user experience	Aggregating everything into a single app is not a solution and that it is a bad design
28 Aug 2026	Chat GPT	Integrating Google Maps and Weather Data.	Produced an idea that we can simply pull those data from google map and weather apis	We changed the idea from building the app with those data locally to pulling and getting the update data without retaining them in our database	Live and update information about google map and weather can be fetched through different apis
29 Aug 2026	Chat GPT	Deployment and Hosting of the app	Got a suggestion that we can host our app on the AWS platform using the $100 free credits	-	AWS platform can handle for both the app's hosting and its secure databases

9. Self-check

☑ Every FR is a user story with a benefit — all 5 FRs in §3 end in a "so that" clause tied to a real pain.

☑ Every FR has acceptance criteria you could actually test — 2 AC per FR, each a number, a state, or a yes/no.

☑ No FR is vague — no bare "fast", "easy", or "user-friendly" anywhere in §3.

☑ NFRs exist, including security / privacy — NFR-5 is a required access-control check.

☑ Every FR traces back to a stated problem — see "Pain these traces to" under each FR, and §7.

☑ Scope and out-of-scope are stated — §2.1 and §6.

☑ Use-case list matches the FR list — 5 use cases, one per FR (diagram added Week 4).

☑ AI log is honest — §8 shows exactly what was asked, drafted, and hand-corrected.

Course rule: you may use AI to draft. You must log it, verify it against a real stakeholder, and be able to explain every FR in the oral exam. Work you cannot explain earns no credit.